Milana Kergourlay

Mechatronics and Robotics Engineering Student
Seeking a 16-Week Assistant Engineer Internship

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Currently in my 2nd year at SeaTech (Master's degree level / Bac+4) specializing in Marine Mechatronics and Underwater Robotics, I am seeking a 16-week international assistant engineer internship starting May 2027. Equipped with solid foundations in kinematics, dynamics, system control, and Linux/ROS environments, I am eager to contribute to innovative robotics projects.

Professional Experience

Marine Engineering Intern – PIRIOU
Concarneau, France *June – July 2026*

- **Field Operations (Workshop & Vessels):** Performed general mechanical maintenance tasks on board ships. Handled technical interventions on marine diesel propulsion engines and auxiliary systems (cooling systems, separators).
- **Technical Office:** Modeled, modified, and updated ship blueprints using **AutoCAD**.

Technical Projects

Port Development Project – Dunkirk
Partnership: Terres Océanes / SeaTech *2026*

- Analyzed wave attenuation performance for three types of high-performance concrete floating breakwater modules.
- Conducted experimental $1/66^{\text{th}}$ scale wave flume testing at SeaTech.
- Calibrated measuring sensors, managed physical model improvements, and authored the scientific report.

Brest Cable Car Modeling
Pre-Engineering Graduate Project *2025*

- Mechanical modeling of the distance compensation system between the upper and lower ropeways.
- Developed a numerical simulation coded in Python (mechanical and energetic engineering concepts).

Leadership & Volunteering

Student Board – SeaTech *Since 2026*

- Created visual assets (Canva) and managed social media channels to engage the student community.
- Coordinated with various student departments to promote university events.

Arts Bureau – University of Toulon *Since 2026*

- Salsa tutor: Planned, organized, and taught weekly dance classes.

Certifications & Interests

- **First Aid:** Certified First Responder (PSE1 & PSE2, France)
- **Credentials:** Valid Driver's License & Personal Vehicle
- **Hobbies:** Salsa dancing, crossfit, swimming, singing, painting

Education

SeaTech School of Engineering – Toulon, France *Since 2025*

- General engineering curriculum with a marine focus: mechanics, energetics, and automatics.
- Specialization : **Master in Mechatronics and Robotic Systems.**
  - **Robotics Core:** Kinematics, dynamics, modeling, and control of terrestrial and marine robotic systems.
  - **Software and Systems:** Development under Linux (Ubuntu) and ROS / ROS 2 environments.
  - **Mechatronics:** Design of smart electromechanical systems, instrumentation, sensor fusion, and control laws.

Intensive Scientific Pre-Engineering Classes (CPGE)
Pôle Kerichen – Vauban, Brest, France *2023 – 2025*

- Two-year intensive preparation in physics and engineering science (PCSI/PSI). Developed analytical and industrial system modeling skills.

General Baccalaureate – High Honors
Lycée Pierre Guéguin, Concarneau, France *2023*

- Specialized in STEM fields.

Computer Skills

- **Robotics and OS:** Linux (Ubuntu), ROS / ROS 2, Matlab / Simulink
- **Programming:** Python, C, C++, Java, SQL
- **CAD and Simulation:** AutoCAD, SolidWorks, 3DExperience
- **Development Tools:** LaTeX, Visual Studio Code, Git, Pyzo

Languages

- **French:** Native speaker
- **English:** B2 (Upper-Intermediate)
- **Ukrainian:** Conversational / Basic knowledge

International Experience

- **Language Stay:** Brighton, UK (2024). Full cultural immersion and intensive daily English practice.